

# STT 465 – Homework 2: Single-Parameter Bayesian Models

Mihir Patel

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## Question 1: Bernoulli-Beta Posterior

[1 point]

A basketball player makes 18 of 25 free throws. Let  $X_i \mid \omega \sim \text{Bern}(\omega)$ ,  $i = 1, \dots, 25$ , where  $\omega$  is the player's underlying probability of making a free throw. Assume the prior  $\omega \sim \text{Beta}(2, 2)$ .

Throughout, let  $n = 25$ ,  $s = \sum_{i=1}^{25} x_i = 18$  (makes), and  $n - s = 7$  (misses). The prior hyperparameters are  $a = b = 2$ .

- (a) Write the likelihood function  $L(\omega \mid x)$  up to a multiplicative constant that does not depend on  $\omega$ . [0.2 pt]

**Solution.** Since the  $X_i$  are conditionally independent given  $\omega$ ,

$$L(\omega \mid x) = \prod_{i=1}^{25} \omega^{x_i} (1 - \omega)^{1-x_i} = \omega^{\sum x_i} (1 - \omega)^{n - \sum x_i} = \omega^{18} (1 - \omega)^7, \quad 0 \leq \omega \leq 1.$$

- (b) Use Bayes' theorem to derive the posterior distribution  $\pi(\omega \mid x)$ . Identify the posterior distribution and its parameters. [0.2 pt]

**Solution.** The  $\text{Beta}(2, 2)$  prior density is

$$\pi(\omega) = \frac{1}{B(2, 2)} \omega^{2-1} (1 - \omega)^{2-1} \propto \omega(1 - \omega).$$

By Bayes' theorem,

$$\pi(\omega \mid x) = \frac{L(\omega \mid x) \pi(\omega)}{\int_0^1 L(\omega \mid x) \pi(\omega) d\omega} \propto \omega^{18} (1 - \omega)^7 \cdot \omega(1 - \omega) = \omega^{19} (1 - \omega)^8 = \omega^{20-1} (1 - \omega)^{9-1}.$$

This is the kernel of a Beta density, so after normalizing,

$$\boxed{\omega \mid x \sim \text{Beta}(20, 9)}, \quad \pi(\omega \mid x) = \frac{1}{B(20, 9)} \omega^{19} (1 - \omega)^8, \quad 0 < \omega < 1.$$

In general, a  $\text{Beta}(a, b)$  prior with  $s$  successes in  $n$  trials gives a  $\text{Beta}(a + s, b + n - s)$  posterior; here  $(2 + 18, 2 + 7) = (20, 9)$ .

- (c) Derive the posterior mean  $E[\omega \mid x]$  and calculate its numerical value. [0.2 pt]

**Solution.** For  $\omega \sim \text{Beta}(\alpha, \beta)$ ,

$$E[\omega] = \int_0^1 \omega \cdot \frac{\omega^{\alpha-1}(1-\omega)^{\beta-1}}{B(\alpha, \beta)} d\omega = \frac{B(\alpha+1, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha+1)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+1)} = \frac{\alpha}{\alpha+\beta},$$

using  $\Gamma(z+1) = z\Gamma(z)$ . With  $\alpha = 20$ ,  $\beta = 9$ ,

$$E[\omega | x] = \frac{a + s}{a + b + n} = \frac{20}{29} \approx 0.6897.$$

This lies between the prior mean  $2/4 = 0.5$  and the sample proportion  $18/25 = 0.72$ , closer to the data because  $n = 25$  outweighs the prior's effective sample size  $a + b = 4$ .

- (d) Calculate the posterior mode and posterior median. You may use `qbeta()` for the median. [0.2 pt]

**Solution.** For  $\alpha, \beta > 1$ , the Beta mode is found by setting the derivative of the log-density to zero:

$$\frac{d}{d\omega} [(\alpha-1)\log\omega + (\beta-1)\log(1-\omega)] = \frac{\alpha-1}{\omega} - \frac{\beta-1}{1-\omega} = 0 \implies \omega_{\text{mode}} = \frac{\alpha-1}{\alpha+\beta-2}.$$

So

$$\text{Mode} = \frac{19}{27} \approx 0.7037.$$

The median has no closed form, so we use the Beta quantile function:

```
qbeta(0.5, shape1 = 20, shape2 = 9)
#> [1] 0.6940678
```

Median  $\approx 0.6941$ .

Mean (0.6897) < median (0.6941) < mode (0.7037), consistent with a slightly left-skewed posterior ( $\alpha > \beta$ ).

- (e) Use R to calculate  $P(\omega > 0.70 | x)$ . Include the R command you use. [0.2 pt]

**Solution.**

$$P(\omega > 0.70 | x) = 1 - F_{\omega|x}(0.70) = \int_{0.70}^1 \frac{\omega^{19}(1-\omega)^8}{B(20, 9)} d\omega.$$

```
1 - pbeta(0.70, shape1 = 20, shape2 = 9)
#> [1] 0.4724805

# Equivalent:
pbeta(0.70, shape1 = 20, shape2 = 9, lower.tail = FALSE)
```

$$P(\omega > 0.70 | x) \approx 0.4725.$$

Given the data and prior, there is about a 47% posterior probability that the player's true free-throw percentage exceeds 70%.

## Question 2: Prior Sensitivity and Bayesian Updating [1 point]

Suppose a treatment is effective for 18 of 21 patients. Let  $\omega$  denote the probability that the treatment is effective. Consider the two priors

$$\text{Prior A: } \omega \sim \text{Beta}(1, 1), \quad \text{Prior B: } \omega \sim \text{Beta}(4, 0.9).$$

- (a) Derive the posterior distribution under Prior A. [0.2 pt]

**Solution.**

- (b) Derive the posterior distribution under Prior B. [0.2 pt]

**Solution.**

- (c) For a general prior  $\omega \sim \text{Beta}(a, b)$ , show that the posterior mean can be written as

$$E[\omega \mid x] = \frac{a + b}{a + b + n} \cdot \frac{a}{a + b} + \frac{n}{a + b + n} \cdot \frac{x}{n}.$$

Explain what the two terms represent. [0.2 pt]

**Solution.**

- (d) Compute the posterior mean under Priors A and B. Briefly explain why the two answers differ. [0.2 pt]

**Solution.**

- (e) Use R to plot the two posterior densities on the same graph. Add a vertical line at the sample proportion  $\tilde{\omega} = 18/21$ . Briefly describe what the plot shows. [0.2 pt]

**Solution.**

### Question 3: Credible Intervals and HPDI

[1 point]

Suppose that after observing data, the posterior distribution is  $\omega \mid x \sim \text{Beta}(22, 4)$ .

- (a) Write the R commands needed to calculate the 95% equal-tailed credible interval  $[F_{\omega|x}^{-1}(0.025), F_{\omega|x}^{-1}(0.975)]$ . Report the numerical interval. [0.2 pt]

**Solution.**

- (b) Explain in one or two sentences the Bayesian interpretation of this 95% credible interval. [0.2 pt]

**Solution.**

- (c) Generate 10,000 posterior samples using the code below. Use the *HDInterval* package to estimate the 95% HPDI. [0.2 pt]

```
set.seed(1)
omega_post <- rbeta(10000, shape1 = 22, shape2 = 4)
```

**Solution.**

- (d) Compare the equal-tailed credible interval and the HPDI. Which one is shorter? [0.2 pt]

**Solution.**

- (e) Explain why the two intervals need not be identical for a skewed posterior distribution. [0.2 pt]

**Solution.**

### Question 4: Comparing Two Population Proportions

[1 point]

A study of births reports that among 980 births in one group, 437 were female. In a second group, among 820 births, 398 were female. Let  $\omega_x$  = female birth proportion in group 1 and  $\omega_y$  = female birth proportion in group 2. Assume independent priors  $\omega_x \sim \text{Beta}(1, 1)$ ,  $\omega_y \sim \text{Beta}(1, 1)$ .

- (a) Derive the posterior distributions of  $\omega_x$  and  $\omega_y$ . [0.2 pt]

**Solution.**

- (b) Calculate the posterior means  $E[\omega_x | x]$  and  $E[\omega_y | y]$ . [0.2 pt]

**Solution.**

- (c) Use the following posterior simulation idea to estimate  $P(\omega_x < \omega_y | x, y)$ . Complete the missing parts of the R code. [0.2 pt]

```
set.seed(1)
U <- 10000
omega_x <- rbeta(U, shape1 = _____, shape2 = _____)
omega_y <- rbeta(U, shape1 = _____, shape2 = _____)
mean(omega_x < omega_y)
```

**Solution.**

- (d) Explain in words what the probability  $P(\omega_x < \omega_y | x, y)$  means. [0.2 pt]

**Solution.**

- (e) A frequentist analysis uses the code below. Briefly explain the difference between the posterior probability in part (c) and the frequentist p-value. [0.2 pt]

```
prop.test(c(437, 398), c(980, 820), alternative = "less")
```

**Solution.**

## Question 5: Poisson-Gamma Model and Prediction [1 point]

Suppose the number of customer complaints received by a company on day  $i$  satisfies  $X_i | \lambda \sim \text{Poi}(\lambda)$ ,  $i = 1, \dots, n$ . Over 20 days, a total of 54 complaints are observed:  $n = 20$ ,  $\sum_{i=1}^{20} x_i = 54$ . Assume the Gamma prior  $\lambda \sim \text{Ga}(a = 2, b = 1)$ , where  $a$  is the shape and  $b$  is the rate.

- (a) Show that the likelihood function has the form  $L(\lambda | x) \propto \lambda^{\sum_i x_i} \exp(-n\lambda)$ . [0.2 pt]

**Solution.**

- (b) Derive the posterior distribution of  $\lambda$  and identify its parameters. [0.2 pt]

**Solution.**

- (c) Calculate the posterior mean  $E[\lambda | x]$  and a 95% equal-tailed credible interval using `qgamma()`. [0.2 pt]

**Solution.**

- (d) Derive the posterior predictive mean for the number of complaints on a future day,  $E[X_f | x]$ . Explain why it is equal to  $E[\lambda | x]$ . [0.2 pt]

**Solution.**

- (e) Use posterior simulation to estimate  $P(X_f \geq 4 | x)$ , using the structure below. Report your estimate and explain what it means. [0.2 pt]

```
set.seed(1)
U <- 10000
lambda_post <- rgamma(U, shape = _____, rate = _____)
x_future <- rpois(U, lambda = lambda_post)
mean(x_future >= 4)
```

**Solution.**